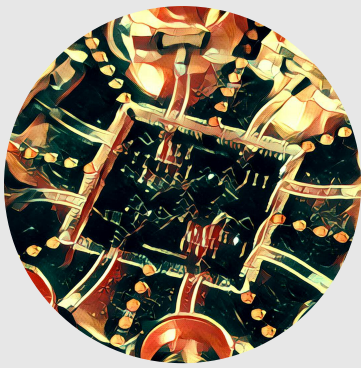




Jorden Senior

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Interests

My current research interests are in the design, fabrication and measurement of quantum phenomena in mesoscopic devices, with a particular interest in superconducting circuits.

My present research focuses on using superconducting qubits and ultra sensitive bolometry to provide a platform for addressing fundamental quantum thermodynamics.

I have an interest in public engagement and outreach activities, and actively participate in a variety of events.

Skills

Competences in:

Python, C++, MATLAB, Java, $\text{\LaTeX}$, Mathematica, Linux, Microsoft Windows, OSX, iWorks, Microsoft Office. Nanofabrication, Cryogenics, Mesoscopic device physics, Microwave engineering, Data acquisition, Measurement automation

education

- 2015–2019 **Ph.D. candidate in Engineering Physics** Aalto University, Finland
Quantum thermodynamics with superconducting circuits
Supervisor: Jukka Pekola
- 2011–2015 **MPhys(Hons) Integrated Masters in Physics** Lancaster University, UK
Towards a Low Frequency Cryogenic Current Amplifier
Supervisor: Jonathan Prance
- 2009–2011 **A Levels** Oldham Sixth Form College, UK
Maths, Further Maths, Chemistry, Physics, French, Critical Thinking, Extended Project Qualification

publications

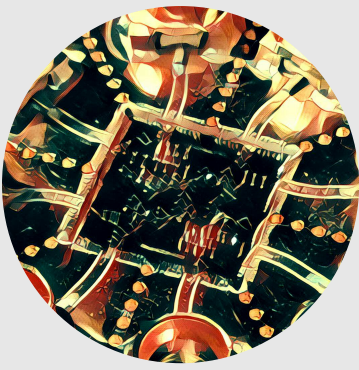
- 2018 *'Realisation of a Quantum Heat Valve'*
A. Ronzani et al, preprint arXiv:1801.09312
- 2017 *'Multiplexing Superconducting Qubit Circuit for Single Microwave Photon Generation'*
R.E. George et al, JLTP, doi:10.1007/s10909-017-1787-x
- 2016 *'Direct measurements of the extraordinary optical momentum and transverse spin-dependent force using a nano-cantilever'*
M.Antognozzi et al, Nature Physics, doi:10.1038/nphys3732
- 2015 Master's Thesis, Lancaster University

awards

- 2015 Ogden Trust 'BLOGden' Blogger of the Year
- 2011–15 Ogden Trust Undergraduate Scholarship
- 2010 British Science Association Gold CREST Award
- 2010 Engineering Education Scheme Certificate awarded by the Engineering Development Trust
- 2009 Duke of Edinburgh Award

experience

- 2014 **Summer Student** University of Bristol, UK
Measuring the light pressure of an evanescent wave using nano-cantilevers, involving the study of optical currents and momenta, both theoretically and experimentally. For this project, I was also nominated for the Ogden Trust Outstanding Intern of the Year. This work was reported in Nature Physics, See doi:10.1038/nphys3732.
- 2013 **Summer Student** Lancaster University, UK
Development of a cryogenic probe to study electronic properties of surface mount electronic devices.
- 2012 **Vice President, LUPAS** Lancaster University, UK
Organising finances, group activities and seminars for Lancaster University physics and astronomy student society
- 2010–2011 **Summer Intern** University of Manchester, UK
Construction of a muon tomography array and analysis of SuperNEMO collaboration 90-cell Geiger detector array.
- 2009–2010 **EES Student** Diodes Semiconductors, UK
Optimisation of diode diagnostic and quality checking protocols. For this project, I also received a British Science Association Gold CREST award.
- 2007–2008 **SCHOME Satellite Proposal** Open University, UK
Research and development of low earth orbit satellite to study spectroscopy, climate, and habitability from Earth using orbital measurement of Earthshine.



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Mathematica, Linux, Microsoft
Windows, OSX, iWorks, Microsoft
Office. Nanofabrication, Cryogenics,
Mesoscopic device physics,
Microwave engineering, Data
acquisition, Measurement automation

Academic Presentations

- 2017 Talk: Flux tunable heat transport across superconducting circuits, LT28, Gothenburg
- 2017 Poster: Flux tunable heat transport across superconducting circuits, QTC 2017, Helsinki
- 2017 Poster: Flux tunable heat transport across superconducting circuits, NanoQI Summer School, San Sebastian
- 2016 Poster: Photon Sourcing with Superconducting Qubits, Microphoton Workshop, London
- 2016 Poster: Microphotonics with Superconducting Qubits, Physics Days, Oulu
- 2015 Poster/Talk: Towards a Low Frequency Current Amplifier, Undergraduate Miniconference, Lancaster University
- 2010 Participant, UK Neutrino Network Meeting, University of Manchester
- 2008 Speaker: Online virtual environments as education platforms with the Open University, Handheld Learning Conference

Outreach and Teaching

- 2015 Co-creator: BSA Gold CREST Project in Superconducting Circuits, Lancaster University (17 - 20 years)
- 2015 Demonstrator, Institute of Physics Outreach, Holster Garden Show
- 2014/15 Co-creator: 'From the very big to the very small', (14 - 18 years)
- 2012-15 Teaching Assistant, Physics Department, Lancaster University
- 2014/15 Co-creator, A Level Workshop on Superconductors, Low Temperature Physics MasterClass, Lancaster University (16 - 18 years)
- 2013/14 Demonstrator, IOP Outreach, Lancaster Science Festival
- 2013/14 Project Supervisor, Superconductivity Project, Physics Residential, Lancaster University (16 - 18 years)
- 2013 Assistant, 'The Power Of...' Public Lecture series, Lancaster University
- 2013 Demonstrator, Big Bang National Science Fair, London

References

PhD studies:

Jukka Pekola [Aalto University, jukka.pekola@aalto.fi](mailto:jukka.pekola@aalto.fi)

MPhys studies:

Jonathan Prance [Lancaster University, j.prance@lancaster.ac.uk](mailto:j.prance@lancaster.ac.uk)

Viktor Tsepelin [Lancaster University, v.tsepelin@lancaster.ac.uk](mailto:v.tsepelin@lancaster.ac.uk)